

Who grades the grader?

APEX-Invariance: understanding what changes when APEX changes its judge

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1. The opportunity

APEX's scores depend on the judgment of the model grading the work. Improving that judge can involve a trade-off. In APEX-Agents 1.1, Mercor improved detection of scattergunned answers while false negatives rose from 5.3% to 8.0% relative to the naive grader [3]. That raises a practical question for the next upgrade: **which kinds of work will the new judge grade more reliably, and where will it introduce errors?**

I propose APEX-Invariance to help the APEX team answer that question. The evaluation would follow individual rubric criteria through controlled changes to the work, identifying which verdicts should remain stable and which should change. It would then connect those results to separate false-approval and false-rejection rates for numeric, semantic and presentation criteria. The outcome would be a reusable regression suite and a report showing where a judge change improves grading, where further review is warranted, and where a deterministic check could help.

Mercor already validates its judges against human labels. APEX-Agents reports 98.5% accuracy on 747 labeled criteria, while APEX-Accounting reports 97.1% accuracy with triple annotation [1, 2]. The proposed method would build on that foundation by testing whether controlled comparisons reveal reproducible failure modes that aggregate accuracy misses. It would not assume that Mercor lacks similar tests internally; the fellowship would begin by identifying the additional evidence its team would find useful.

The research contribution

The project would investigate two mechanisms behind unreliable verdicts. First, an upstream error may affect several rubric criteria even when their declared dependencies do not capture that propagation. Second, a numeric checker may improve grading only if it can identify the right quantity, unit and conditions before applying a tolerance rule. The central comparison is therefore empirical: does diagnosing these mechanisms lead to better grading or lower review cost than selecting a judge on held-out accuracy alone?

CheckList provides a framework for behavioral testing [5], while Chen et al. distinguish verdicts that should remain stable from those that should change [4]. Rubric-edit studies show why both matter [7, 8], and MBABench already validates a finance judge with seeded errors [10]. Building on these precedents, this project would test whether tracing cross-criterion propagation and validating quantity alignment improve grading decisions on held-out APEX work. Expert labels and paired comparisons make that incremental claim testable.

Preparation and fellowship scope

I have completed an audit of 100 public tasks, independently reproduced 19 numeric criteria, collected 2,394 valid judgments across four local studies, and implemented a conservative acceptance-bound procedure. The local studies use constructed controls and establish useful test cases; they do not establish failures in Mercor's reference judge.

The proposed fellowship would supply the next layer of evidence. I would work with Mercor to obtain the reference configuration, authorized agent responses and support from its professional evaluators. These resources would make independent validation on professional work possible.

2. What the preliminary evidence shows

The public-data audit covers 100 tasks and 1,140 criteria at a pinned revision of APEX-v1-extended [13], the development release accompanying the original APEX report [14]. I reproduced 19 numeric criteria from source files across five tasks. The experiments below are my preliminary studies, summarized in Appendix A; their reference calculations come from the published task files.

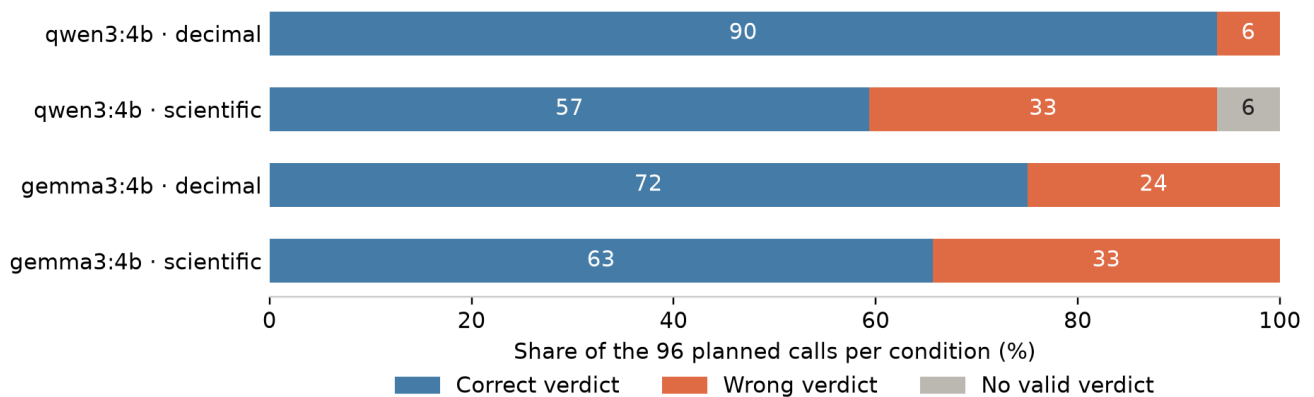
Equal accuracy can conceal different consequences

Four local judges each made 12 errors in 54 English numeric judgments, but their error profiles differed. Qwen 1.7B approved 12 of 21 incorrect-value judgments and rejected none of the 33 acceptable ones; Qwen 8B made no false approvals but rejected 12 of the 33 acceptable judgments. A separate study found that Gemma 3 4B returned the same verdict on every repeat in all 96 test conditions while falsely approving 57 of 144 judgments on incorrect values.

Equivalent numbers can receive different verdicts

The latest study paired decimal and scientific notation for 32 criterion-value combinations across three public task families. Qwen changed its verdict in 9 of 30 complete pairs and Gemma in 9 of 32. Both judges returned stable verdicts when the same prompt was repeated. For example, Qwen accepted the Gen Z NPS value -28 in all three repeats but rejected the equivalent $-2.8E+1$ in all three.

Equivalent notation, different grading outcomes



My local experiment included 96 planned calls per condition. Numbers inside bars show counts; missing verdicts are reported separately from errors.

The study produced 378 valid verdicts from 384 planned requests. Six Qwen requests remained truncated after one unchanged retry. The resulting 12 failed attempts are preserved and excluded from grading-error counts. Eighteen of the 64 distinct prompts had appeared in earlier studies, so this remains a development diagnostic, not a held-out evaluation.

Correct arithmetic is only part of the problem

A restricted parser correctly checked all 64 clear numeric test cases and abstained on 36 syntax controls. Yet it approved all six responses that reported the same value for the wrong named target. Separately, omitting one cash flow in task 2287 moved all 11 numeric criteria outside tolerance despite zero declared dependency edges. These findings motivate testing quantity alignment and error propagation; neither establishes performance on natural professional responses. Independent professional review would be part of the fellowship.

3. How the evaluation would work

Each response would be compared with a controlled variant whose expected effect is established before judging. Meaning-preserving changes should preserve verdicts; substantive changes should affect only the labels validated as changed. These paired tests apply the invariance and sensitivity framework of Chen et al. [4] to professional rubrics.

Intervention	Expected effect	Control
Equivalent numeric format or criterion paraphrase	Preserve the correct verdict.	Same response and criterion, repeated without an edit.
Move a value outside its published tolerance	Reverse an otherwise acceptable verdict.	Matched value inside the interval, including inclusive endpoints.
Revise a requirement [8]	Change only the response-criterion labels affected by the revision.	A response whose correct label is unaffected.
Seed an upstream calculation error [10]	Change the labels of verified affected criteria.	The unseeded response and criteria outside the verified effect.

Ground the labels before measuring the judge

Following the annotation precedents in APEX-Accounting and Chen et al. [2, 4], three Mercor experts would independently label each selected response and identify the supporting evidence. Adjudication would preserve their original judgments so that ambiguity and disagreement remain visible.

Numeric reference labels would use published tolerances and calculations reproduced from source files. Experts would also verify the entity, metric, period and conditions. Unresolved interpretations would be excluded under a rule fixed before judging and counted in the ambiguity report.

Each criterion would receive a primary scoring class: numeric with tolerance, semantic content, or presentation and structure. Mixed criteria would retain secondary tags. A **task family** would include all criteria, responses, transformations and repeats from one task. Each family would belong entirely to either the calibration set or the test set. This prevents variants of the same problem from appearing on both sides of the split.

Test whether the diagnosis changes a useful decision

The baseline would select a judge by aggregate accuracy on labeled natural calibration responses. The proposed workflow would use that same set plus controlled tests to choose class-specific policies, including the numeric intervention. Additional annotation and computation would be counted in its cost. All prompts, extraction rules and policy choices would be fixed before testing.

Both workflows would be evaluated on the same natural responses for false approvals, false rejections, automated coverage and review cost. A separate report on the controlled variants would explain the failures behind those decisions.

Evaluate the numeric intervention as a complete system

The checker would verify the extracted value's quantity, unit and conditions before applying the numeric rule, abstaining when alignment is uncertain. Selective judging motivates reporting error alongside coverage [6]; here, the checker and LLM would be compared on the same selected cells, with review cost also reported. This comparison would show whether the checker improves grading on the cases it handles. The local failures on responses naming the wrong target show why quantity alignment needs its own evaluation.

4. Analysis, uncertainty and recommendations

The primary report would separate two errors with different consequences. A false approval is a “met” verdict on an expert-labeled unmet criterion; its denominator is unmet criteria. A false rejection is a “not met” verdict on an expert-labeled met criterion; its denominator is met criteria. Each rate would be computed within a family and then averaged equally across eligible families, with family and label counts reported. Criterion-weighted rates would provide a secondary view.

For a transformation, the primary contrast would be paired verdict disagreement minus same-condition repeat disagreement. Propagation analysis would distinguish expert-verified label changes, correct judge reactions and spurious verdict changes. The directly injected error would not count as a downstream effect. Exploratory intervals would use a cluster bootstrap [16], resampling whole families rather than individual criterion calls. Balanced accuracy [9], expert agreement, ambiguity and exclusions would complement the primary outcomes.

A frozen judge configuration would include the prompt, context, repeat count, aggregation and failure handling. Diagnostic repetitions would be reported explicitly; a truncated or invalid output would remain missing rather than become a failed criterion. Comparisons would show the unresolved count and the range of results allowed by those missing verdicts.

Distinguish a useful diagnosis from an acceptance guarantee

The core study would compare grading methods and make recommendations that reflect the available precision. Proposed strict acceptance limits are 2% false approvals and 5% false rejections per class, subject to agreement with Mercor before testing. Automatic-grading recommendations would require valid uncertainty bounds below both limits. A small observed error rate alone would not qualify.

A conservative procedure is already implemented. Within each eligible family, the error fraction cannot exceed an indicator of whether that family contains any error. An exact one-sided binomial upper bound on this family-event probability therefore also bounds the equal-family mean error. The construction uses Clopper–Pearson inversion [12], with a 0.05/6 allocation across three classes and two error types for one configuration selected on calibration data. Unlike applying a binomial interval to pooled criterion calls, it allows arbitrary dependence within a family.

The guarantee still requires independent eligible-family sampling, correct labels, a fixed audit size and a frozen configuration. Convenience samples or pooled fixed quotas do not establish that sampling assumption. Offering acceptance for any of four test-evaluated configurations would require adjustment across 24 bounds or a fresh independent audit. This is a different target from the error conditional on acceptance studied by Badshah et al. [6].

What the planned sample can support

Even with zero errors in 45 eligible test families, the six-comparison upper bound is 10.09%. Under this conservative procedure, a 2% bound would require at least 237 independent eligible families with zero errors; a 5% bound would require 94. These are sufficient sample counts for this construction, not universal lower bounds.

The proposed six-month budget therefore funds diagnosis, regression comparisons and a test of the intervention. A strict automatic-acceptance guarantee would require a separately scoped audit, different limits agreed before testing, or a justified sharper procedure. An implementation study of 24 simulated scenarios passed its pre-specified coverage check; those simulations are not empirical evidence about APEX and do not establish the real-world sampling assumptions.

5. Workplan, resources and value to Mercor

Establish the reference and validate the pilot

In the first two weeks, I would work with an APEX owner to confirm the reference configuration, understand existing tools and identify the decision that needs additional evidence. I would then adapt the local suite, retaining passed controls as regression tests and focusing new annotation on remaining gaps.

In weeks three through six, I would pilot up to five public task families within 40 expert hours. Expansion would require at least four families with adjudicated labels, fewer than 15% ambiguous criteria and a measured cost within budget. The propagation test would remain incomplete unless all five families were evaluated.

Expand only after the pilot supports it

The six-month target is 60 new consulting and investment-banking families; the three-month option targets 24. Four frontier agents would produce responses. I would select three responses per family through a pre-registered balanced rotation, without using judge scores to choose them. Separate experts would author tasks, solve them independently and grade responses. Four judge configurations and the numeric intervention would be evaluated on held-out families.

Planning assumption	Three months	Six months
New families, calibration / test	24, split 8 / 16	60, split 15 / 45
Expert hours, including pilot and reserve	384	780
Expert budget at USD 100/hour plus 20% contingency	USD 46,080	USD 93,600

Each new family is budgeted for eight hours of construction and independent re-solve, plus three hours of grading: three responses, three graders and twenty minutes each. Totals include the 40-hour pilot and an 80-hour adjudication reserve. These are planning assumptions, not quoted Mercor rates. At ten criteria per response and three judge repetitions, the six-month comparison would require 21,600 core criterion calls. Agent generation, controls, extraction, API charges, any separate acceptance audit and the fellow stipend are additional.

Deliver an instrument the team can reuse

Mercor would receive an expert-reviewed dataset, a regression suite with known expected effects, and a report linking each scoring class's errors to an intervention or review recommendation. If error is not concentrated by class, or the checker does not improve the paired comparison, those results would be reported directly. If equivalent tools already exist internally, their remaining limitations would determine the study's scope. Publication and release of examples would follow Mercor's data-access terms; the held-out package could remain private.

A separately scoped multilingual and multicultural extension could begin with Spanish and Latin American professional contexts, motivated by prior evidence of multilingual judge inconsistency [15]. Experts would validate equivalent translations to test grading transfer, while tasks adapted to local professional requirements would need new labels. This would distinguish language effects from changes in what counts as acceptable work, giving Mercor evidence to assess opportunities in new evaluation markets. Demand and commercial value would need separate validation. At 60 families, an illustrative seven additional expert hours per family would add USD 50,400 under the same rate and contingency assumptions; this extension is outside the core budget.

I would lead measurement design, implementation and analysis. As first author of MeSHClass-ES and AnatEM-ES (BioNLP 2026), I released Spanish biomedical resources with reproducible translation and evaluation pipelines [11]. My preparation for this proposal applies that experience to APEX; Mercor's evaluators would provide the independent professional validation needed during the fellowship. I am based in Bogotá, available full time for six months, and willing to relocate if needed.

Appendix A. Research tests and evidence record

I recorded these thresholds before collecting reference-judge results or independent expert judgments. The identifiers preserve the research record. These proposed fellowship tests do not replace the acceptance procedure in Section 4.

Test	Prediction and evidence against it
H8: concentration by class	At least one class has an error rate at least twice the global rate for the same error type. The claim is refuted if every class is within 1.5 times the global rate for both error types. Ratios between 1.5 and 2 are inconclusive, and ratios are undefined when the global error rate is zero. Absolute rates and uncertainty would accompany every comparison.
H4: numeric boundaries	Reference-judge error on inclusive endpoints and just-inside values exceeds 5%. The refutation condition is error at or below 5% with an interval upper bound at or below 10%. This does not imply acceptance at a stricter limit.
H5b: propagation	In at least one of five pilot families, an error changes an expert-verified label outside the declared dependents, and the judge follows that change. Refutation requires all verified propagated changes to fall within declared descendant sets in all five families. This concerns tested interventions, not graph completeness or annotation defects.
H6: numeric intervention	After quantity alignment and extraction, the checker lowers error by at least two percentage points at coverage of at least 70% on held-out families. A gain below two points or coverage below 70% refutes the claim. Both methods would be compared on the same selected cases, with review cost reported.
H7: feasibility	Task construction and independent solving take at most eight expert hours per family. Grading takes at most twenty minutes per response per grader. Pilot medians above twelve hours or thirty minutes refute those feasibility assumptions. Intermediate costs require budget revision.
H2 / H3: optional language transfer	Equivalent Spanish responses or Spanish grading add at least three percentage points of disagreement beyond same-condition repeats. A 95% family-bootstrap interval entirely below that threshold refutes the effect; an interval crossing it is inconclusive. Small disagreement does not establish acceptable absolute error.

Completed local studies

Study	Valid judgments	Scope and result
Four-model comparison	1,080	Equal total error concealed opposing false-approval and false-rejection profiles.
Numeric prompt intervention	360	A fixed clarification improved one of thirty Qwen majority cells and did not generalize.
Content and paraphrase	576	Stable repeated verdicts coexisted with false approvals.
Boundary and notation	378 / 384 planned	Equivalent notation changed complete paired verdicts in both judges; six Qwen requests remained unresolved.
Total	2,394	The total includes repeats and excludes the separate 108-call exploratory pilot, failed attempts, parser checks and simulations.

The initial English/Spanish split did not persist across three other judges, and prompt clarification failed as a general remedy. Both hypotheses are recorded as refuted. Frozen requests, configurations, source hashes, model identities and raw outputs preserve the evidence.

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